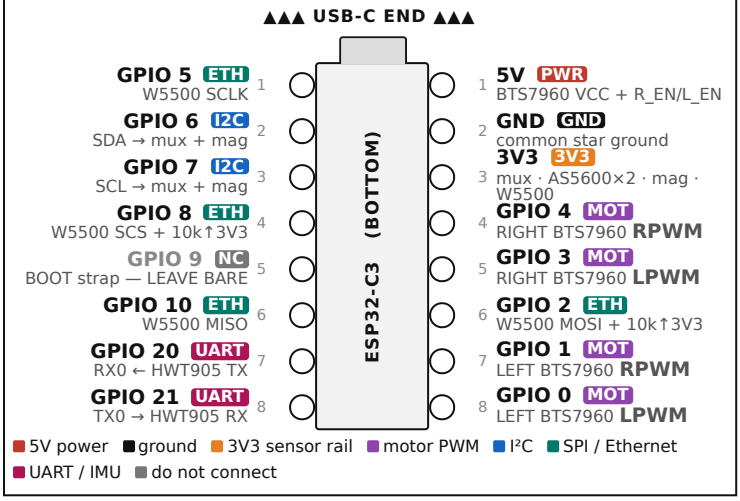
Autonomous Car — Solder Pinout Sheet

ESP32-C3 SuperMini · 2× BTS7960 · 2× AS5600 behind PCA9548A · QMC5883L · W5500 Ethernet · HWT905 IMU

Rev 2026-08-30
from SETUP.md (verified 2026-08-23)
Print at 100% — do not scale

⚠ **BOARD DIAGRAM IS MIRRORED (SOLDER SIDE / BOTTOM VIEW) —** pads are drawn as seen through the back of the board, USB-C at top. Top view is at bottom of page for cross-checking.

ESP32-C3 SuperMini — SOLDER SIDE (flipped over, USB-C away from you at the top)



CROSS-CHECK — TOP VIEW (component side, USB-C at top). Left/right columns are swapped vs. the mirrored diagram above.

LEFT column, top→bottom			RIGHT column, top→bottom		
1	5V	BTS VCC	1	IO5	ETH SCLK
2	GND	ground	2	IO6	I2C SDA
3	3V3	sensors	3	IO7	I2C SCL
4	IO4	R RPWM	4	IO8	ETH SCS
5	IO3	R LPWM	5	IO9	— NC —
6	IO2	ETH MOSI	6	IO10	ETH MISO
7	IO1	L RPWM	7	IO20	IMU RX0
8	IO0	L LPWM	8	IO21	IMU TX0

Before you power up — check list

- **GPIO 9 must stay bare.** If anything holds it low at reset the C3 boots into download mode instead of running. This is why W5500 MISO is on GPIO 10.
- **GPIO 2 and GPIO 8 each need a 10k pull-up to 3V3.** Both are boot-strapping pins that must read HIGH at reset. On GPIO 8 the pull-up doubles as the correct idle-high state for SPI chip-select. Expect the onboard blue LED (GPIO 8) to flicker on Ethernet traffic — that is normal, not a fault.
- **Ground first, then signals, then power.** Motor ground and logic ground must meet at one point at the battery negative, not daisy-chained through the ESP32.
- **Never drive RPWM and LPWM of the same BTS7960 high outside the firmware's brake state.** Both low = coast, both high = brake.
- Continuity-check every pad against this sheet *before* the first power-up. Ohm 5V↔GND and 3V3↔GND for shorts.

Unused / reserved

GPIO 9	BOOT strapping pin — **leave completely unconnected**
ESP32 WiFi / BT	Off at boot by design — they preempted the 500 Hz control loop. Bring WiFi up on demand with A 1. The long-term plan is a separate 2.4 GHz link; no pins are reserved for it here.

1 Power & ground

From		To
Battery +	→	BTS7960 L B+ · BTS7960 R B+ *(via main switch + fuse)*
Battery −	→	BTS7960 L B− · BTS7960 R B− · **star ground point**
5V source	→	ESP32 5V pad, BTS7960 L+R VCC, R_EN, L_EN
ESP32 3V3	→	PCA9548A VCC, AS5600 L VCC, AS5600 R VCC, GY-271 VCC, W5500 3V3, HWT905 VCC
ESP32 GND	→	star ground — and every module GND back to that same point

Not recorded in SETUP.md: battery chemistry/voltage and where the 5 V rail comes from (USB vs. an onboard buck). **Confirm both before soldering B+.** If 5 V comes only from USB, the BTS7960 logic rail is unpowered when running on battery.

2 Motor drivers — 2× BTS7960 (8-pin logic header)

BTS7960 pin	LEFT module	RIGHT module
RPWM	**GPIO 1**	**GPIO 4**
LPWM	**GPIO 0**	**GPIO 3**
R_EN	5V (strapped)	5V (strapped)
L_EN	5V (strapped)	5V (strapped)
R_IS / L_IS	not connected	not connected
VCC / GND	5V / star GND	5V / star GND
M+ / M−	LEFT motor	RIGHT motor
B+ / B−	battery	battery

Motors are mounted **mirror-imaged and wired identically** — do **not** swap M+/M− on one side to "fix" direction. The sign flip lives in firmware and in `calibration.json` (`wheels.LEFT.direction = −1`). Rewiring it here breaks the calibration.

3 I²C bus — SDA GPIO 6, SCL GPIO 7

Device	Addr	Wiring
PCA9548A (HW-617)	0x70	SDA→GPIO 6, SCL→GPIO 7, VCC→3V3, GND, A0/A1/A2→GND, RST→3V3
GY-271 compass *(really QMC5883L)*	0x0D	**Main bus, ahead of the mux.** SDA→GPIO 6, SCL→GPIO 7, VCC→3V3, GND. DRDY not connected.
AS5600 LEFT	0x36	Mux **channel 6**: SD6→SDA, SC6→SCL. VCC→3V3, GND, DIR→GND.
AS5600 RIGHT	0x36	Mux **channel 3**: SD3→SDA, SC3→SCL. VCC→3V3, GND, DIR→GND.

Both encoders share 0x36 — that is the entire reason the mux exists. Keep **one** set of 4.7k pull-ups on the main bus (the mux board usually has them); do not stack a second set from the compass board. Encoder magnet airgap is already marginal (AGC pegged at 128, target 64) — do not add spacing while remounting.

4 Ethernet — W5500 NEW · NOT YET WIRED

W5500 pin		ESP32-C3
SCLK / SCK	→	**GPIO 5**
MISO	→	**GPIO 10** *(deliberately NOT GPIO 9)*
MOSI	→	**GPIO 2** + 10k to 3V3
SCS / CS	→	**GPIO 8** + 10k to 3V3
RST	→	10k to 3V3, 100nF to GND — **no GPIO**
INT	→	not connected (polled)
3V3 / GND	→	3V3 rail / star GND + *100nF & 10µF at the module*

This block is a proposal, not as-built — SETUP.md records the W5500 as unwired. It is the only assignment that leaves UART0 free for the IMU. **GPIO 9 is the C3 BOOT strap: a W5500 holding it low at reset drops the board into download mode instead of booting.** Use a 3.3 V-logic W5500 module (W5500 Lite); the 5 V "Ethernet Shield" variants need level shifting. Firmware support does not exist yet.

5 IMU — HWT905-TTL NEW · NOT YET WIRED

HWT905 pin		ESP32-C3
TX	→	**GPIO 20** (UART0 RX) — the only strictly required signal
RX	→	**GPIO 21** (UART0 TX) — optional, config only
VCC / GND	→	3V3 / star GND

115200 baud, 100 Hz. UART0 is free because `ARDUINO_USB_CDC_ON_BOOT=1` puts the console on native USB. **Re-run the on-chip accel calibration after the sensor is rigidly mounted** — the 2026-08-26 bench calibration defined the desk as level and does not transfer to the car. Trust the raw 0x51 accel packet, not the 0x53 euler roll/pitch (fixed +1.16°/−1.17° offset).

Source: `~/autonomus car/SETUP.md` §2, §5a, §5b, §7 · firmware `~/drive-harness-diag/c3-webgui` · USB serial `/dev/ttyACM0` @ 115200

Sections 4 & 5 are proposed, not as-built — verify before soldering.